VARIATIONAL ANCHORS FOR APÉRY-LIKE FAMILIES: CURVE-BLINDNESS AND THE JET CALCULUS OF CELLULAR INTEGRALS

Abstract. For the Brown–Zudilin cellular family for $\zeta(5)$, with cell $T(n,k,l) = \binom{n+k}{n}\binom{n}{k}^2\binom{n+l}{n}\binom{n}{l}^2\binom{n+k+l}{n}$ and rows $(Q_n, \hat{P}_n, P_n)$, we study ε-deformations of the double sum $\sum_{k,l} T$ along polynomial curves in the argument space and prove a structural dichotomy. A short *coupling lemma* shows that the top-graded harmonic content of the ε^r coefficient of any polynomial-curve deformation is a function of the curve's first-order direction alone, through its pure r-th powers; combined with a finite two-prime rank computation this yields *curve-blindness*: pinned curve deformations of any degree have their ε^5 row confined to $\mathbb{Q}Q \oplus \mathbb{Q}\hat{P}$ — the top row P_n is unreachable — although the same deformations generate the full $\zeta(3)$-level structure. The proof identifies the escape: objects whose fifth-order jet is decoupled from their lower jets, analytically the mixed fifth parameter-derivatives of the cellular integral. We then verify the first rung of the resulting *variational tower*: the n-direction first variation $\dot{Q}(n) = \sum_{k,l} T \cdot \Lambda_1$ satisfies exactly the n-differentiated Brown–Zudilin recurrence $L_{\mathrm{BZ}}(\dot{Q})(n) = -\sum_i c_i'(n)\, Q(n+i)$, while the k- and l-direction variations vanish identically (they are gauge). We describe the resulting programme for a certificate-free proof of the weight-five attachment $\sum T \cdot B_5 = \frac{33}{4} P_n$, and three further applications of the same instrument: a validated companion-discovery scanner that rediscovers the Franel and **D** companions from first principles; a two-prime reachability dichotomy across the fifteen sporadic Apéry-like pairs aligned with character and convergence; and per-summand annihilation laws that show the fitting alphabets of the "formula-less" families are degenerate. Every statement carries its evidence label; no finite computation is reported as a proof.

Contents

1. Introduction

1.1. **Context.** The Brown–Zudilin cellular integrals [1] produce the best-known simultaneous rational approximations to $\zeta(3)$ and $\zeta(5)$: an integral family I_n with

$$I_n \;=\; Q_n\,(2\zeta(5) + 4\zeta(3)\zeta(2)) \;-\; 4\hat{P}_n\,\zeta(2) \;-\; 2P_n, \tag{1}$$

whose coefficient rows $(Q_n, \hat{P}_n, P_n)$ satisfy one certified third-order recurrence L_{BZ}. The programme surrounding this paper studies the *companion problem*: closed harmonic-weight formulas $\sum_{k,l} T(n,k,l)\,\mathfrak{w}(n,k,l)$ for the rows, where $T(n,k,l) = \binom{n+k}{n}\binom{n}{k}^2\binom{n+l}{n}\binom{n}{l}^2\binom{n+k+l}{n}$ and $\mathfrak{w}$ is a polynomial in the harmonic letters $H_x^{(r)}$, $x \in \{n, k, l, n \pm k, n \pm l, k + l, n + k + l\}$. The middle row

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is a theorem ($\hat{P}_n = \sum T\,\hat{w}_3$, by the descent of [1] together with a constructive residue calculus); the top row

$$P_n \;=\; \sum_{k,l=0}^{n} T(n,k,l)\,\omega_5(n,k,l) \tag{2}$$

is verified far beyond doubt and is [OPEN] — it is the single identity ("the bridge") carrying the sharp low-prime denominator bound $2^2\cdot 3$ for P_n.

An ε-deformation machinery discovered the compact weights as Taylor coefficients: there is an explicit formal deformation ("family 1") of the summand with

$$\textstyle\sum_{k,l} T\cdot B_3 = \hat{P}_n, \qquad \sum_{k,l} T\cdot B_5 = \frac{33}{4}\,P_n \qquad \text{[VERIFIED: exact } \mathbb{Q},\ n\le 16\text{; mod two primes, } n\le 80\text{]}, \tag{3}$$

where B_m are Bell coefficients of the deformation. Neither identity in (3) is proved; the present paper is about *why* they have resisted proof, and what instrument the resistance itself identifies.

1.2. **What this paper does.**

(1) We define *curve atoms*: deformations $T(n+\alpha(\varepsilon), k+\beta(\varepsilon), l+\gamma(\varepsilon))$ along polynomial curves. These are the meromorphically honest deformations — each is the Brown–Zudilin parametric integrand at shifted parameters.

(2) We prove the *coupling lemma* (Lemma 2.2): the top-graded content of $[\varepsilon^r]$ of a curve atom depends only on the first-order direction, through its pure r-th powers, and the polarization identity is blocked for mixtures (Lemma 2.3).

(3) Combining with finite two-prime rank computations we obtain *curve-blindness* (Theorem 3.1): pinned curve combinations of degree ≤ 3 have ε^5 row confined to $\mathbb{Q}Q \oplus \mathbb{Q}\hat{P}$. The graded structure of the phenomenon — weight ≤ 3 reachable, weight 5 not — matches the motive's weight spike $\{0,3,5\}$.

(4) The proof mechanism identifies the escape (§4): decoupled fifth jets, i.e. mixed fifth parameter-derivatives of the family. We verify the first rung of the corresponding *variational tower* (Theorem 4.1, [VERIFIED: exact $\mathbb{Q}$, $n\le 18$], with an identified proof route): the n-direction first variation satisfies exactly the differentiated recurrence, while the k-, l-directions are gauge.

(5) Applications (§5): the same jet calculus, run as a discovery instrument, (i) rediscovers the Franel and **D** companions from linear atoms with exact held-out validation, producing new null-equivalent representations; (ii) establishes a two-prime reachability dichotomy across the fifteen sporadic pairs aligned with character and convergence; (iii) closes, as completed negatives, the three full-degree fits left unrun by the sporadic-pairs status paper; and (iv) exhibits per-summand annihilation laws (for the double-sum family ζ) showing that fitting alphabets can be degenerate.

1.3. **Evidence discipline.** [THM]/[PROVED] mean complete mathematical proof (here: short and human-readable); [VERIFIED] means exact rational or two-prime modular computation over a stated range; [EXCLUDED] means a proved or two-prime-computational negative; [OPEN] is open. Finite computations are never reported as proofs. Scripts and logs for every [VERIFIED]/[EXCLUDED] statement are listed in §7.

2. Curve atoms and the graded expansion

2.1. **Letters and cells.** Write the cell as a ratio of factorials $T = \prod_L (x_L!)^{p_L}$ over the nine letters

$$(x_L) \;=\; (n,\ k,\ l,\ n{+}k,\ n{+}l,\ n{-}k,\ n{-}l,\ k{+}l,\ n{+}k{+}l), \qquad (p_L) \;=\; (1,\ -3,\ -3,\ 1,\ 1,\ -2,\ -2,\ -1,\ 1).$$

Definition 2.1 (curve atom)**.** A *curve atom of degree D* is a tuple $u=(u_1,\dots,u_D)$, $u_i\in\mathbb{Z}^3$; it denotes the deformed cell $T\big(n+\alpha(\varepsilon),\ k+\beta(\varepsilon),\ l+\gamma(\varepsilon)\big)$ with $(\alpha,\beta,\gamma)(\varepsilon)=\sum_i u_i\varepsilon^i$, each factorial replaced by the corresponding Γ-value. Per letter the shift series is $s_L(\varepsilon) = \sum_i d_{i,L}\,\varepsilon^i$ where $d_{i,L}$ is the letter's linear form evaluated on u_i.

Expanding $\log\Gamma$, the cell equals $T \cdot \exp\bigl(\sum_{m\ge1} \varepsilon^m \Lambda_m\bigr)$ with

$$\Lambda_m \;=\; \sum_L p_L \sum_{j\ge1} \frac{(-1)^{j-1}}{j} \Bigl(H^{(j)}_{x_L} - \zeta(j)\mathbf{1}_{j\ge2}\Bigr)\,[\varepsilon^m]\, s_L^j. \tag{4}$$

The ζ-constants organise into graded blocks; the linear forms $S_i = \sum_L p_L d_{i,L}$ vanish identically for every atom (a two-line check on the letter table), so no Euler constant enters. The $[\varepsilon^r]$ coefficient of the atom is T times a polynomial in harmonic letters and ζ-values; its *harmonic weight* grading assigns $H^{(j)}_x$ weight j and $\zeta(j)$ weight j.

2.2. The coupling lemma.

Lemma 2.2 (coupling; Veronese structure of the top grade)**.** [PROVED] *Fix a curve atom and $r \ge 1$. Every harmonic monomial of weight w occurring in $[\varepsilon^r]$ arises from letter factors with data $[\varepsilon^{m_i}]s_L^{j_i}$, $\sum j_i = w$, $\sum m_i = r$, and always $m_i \ge j_i$. In the top-graded case $w = r$ each factor has $m_i = j_i$, and $[\varepsilon^j]s_L^j = d_{1,L}^j$. Consequently the weight-r component of $[\varepsilon^r]$ of a curve atom depends only on the first-order direction u_1, through the pure powers $d_1^{\otimes r}$; higher curve coefficients $u_2, u_3, \ldots$ contribute only to weights $< r$.*

Proof. $s_L = d_{1,L}\varepsilon + O(\varepsilon^2)$, so $[\varepsilon^m]s_L^j = 0$ for $m < j$ and $= d_{1,L}^j$ for $m = j$. Summing the constraints $m_i \ge j_i$ against $\sum m_i = r = \sum j_i = w$ forces equality factorwise. □

Lemma 2.3 (polarization is blocked for mixtures)**.** [PROVED] *Let a $\mathbb{Q}$-combination of curve atoms have aggregate moments $\mu_m = \sum_v c_v\, d_{1,v}^{\otimes m}$. Then any motion of μ_5 achievable by perturbing directions and coefficients moves μ_4 at the same order: for d, δ directions and $c \in \mathbb{Q}$, $c\bigl[(d+\delta)^{\otimes 5} - d^{\otimes 5}\bigr] = 5c\,\delta{\odot}d^{\otimes 4} + O(c\delta^2)$ while the same perturbation moves μ_4 by $4c\,\delta{\odot}d^{\otimes 3} + O(c\delta^2)$. In particular the fifth-symmetric slot cannot be steered within the Veronese mixture class while all lower moment aggregates are held fixed, in contrast to the free symmetric algebra where polarization decouples the slots.*

Proof. Immediate from the binomial expansion; the point is only that both leading corrections are linear in $c\,\delta$, so no scaling regime separates them. □

Remark 2.4. Lemma 2.3 is where curves differ from formal deformations. The formal ε-machinery assigns each letter an independent data vector $(e_1, \ldots, e_5)$ — the free symmetric algebra — and the known P-emitting deformation (family 1) uses that freedom essentially: its letter l has $e_1 = 6$ while $e_5 \ne 6^5 \cdot$ (const); it is not curve-consistent. The same fact appears on the analytic side as a non-realizability statement: the family-1 letter table admits no Γ-product realization with shift scale ε/N for any $N < 120$, and at $N = 120$ only with multiplicities of order 10^{11} [VERIFIED: integer moment-lattice computation; `eps40_gate.py`]. Decoupled jets have no polynomial-curve realization; that is one fact seen from two sides.

3. Curve-blindness

Fix the pinning conditions on a $\mathbb{Q}$-combination $F(\varepsilon)$ of curve atoms: $[\varepsilon^1]F = [\varepsilon^2]F = 0$ and $[\varepsilon^3]F \in \mathbb{Q}Q \oplus \mathbb{Q}\hat{P}$, imposed separately on every ζ-graded component (with or without a corresponding condition at ε^4).

Theorem 3.1 (curve-blindness)**.** *For curve atoms of degree ≤ 3 with directions in the boxes $u_1 \in \{-2..2\}^3$, $u_2 \in \{-1, 0, 1\}^3$, $u_3 \in \{0, \pm e_i\}$, every pinned $\mathbb{Q}$-combination has*

$$[\varepsilon^5]F \;\in\; \mathbb{Q}Q \;\oplus\; \mathbb{Q}\,\hat{P},$$

i.e. the P_n-component vanishes, in the rational part and in every ζ-graded component. The ε^3 coefficient meanwhile realises $\hat{P}$ with a free coefficient, and quadratic/cubic curves realise $\hat{P}$ at ε^5 as well. Structure of the proof: Lemmas 2.2–2.3 reduce the statement to the finite claim that the pinned Veronese moment fibre's weight-five row image lies in $\mathbb{Q}Q \oplus \mathbb{Q}\hat{P}$; that finite claim is [VERIFIED: two primes $4194301, 4194247$, identical ranks; $n \le 24$; `eps41.py`, `eps42.py`] *at ranks* $53/36$ *(linear),* $88/59$ *(quadratic),* $80/54$ *(cubic) with nullspace dimensions* 99–5075.

Remark 3.2 (reading). The pinning enslaves the top-weight data to the low-order null sector (Lemma 2.2: only u_1 feeds weight five; the graded pinning constrains precisely the u_1-moment aggregates; Lemma 2.3: no escape by polarization), and the null sector's fifth powers are row-blind to P. The graded visibility — curves see the $\zeta(3)$-level structure at every order, never the $\zeta(5)$ level — matches the family's weight spike $\{0, 3, 5\}$: tangential (curve) data computes the subtop graded pieces; the top piece requires a decoupled jet. We regard the theorem's finite part as of the same epistemic kind as a certified rank computation; the conceptual content is the coupling mechanism.

4. The tool: variational anchors

Lemma 2.2 does not merely forbid; it prescribes. To reach the top row one needs objects whose fifth-order data is *decoupled* from their lower jets. Analytically such objects exist and are classical: mixed parameter-derivatives $\partial_{v_1} \cdots \partial_{v_5} I(a)$ of the cellular integral — contour integrals with logarithmic insertions, with no $\varepsilon^{1..4}$ content, spanning the full polarized letter algebra. What they need is an *anchor*: an evaluation principle replacing (1), which is an arithmetic statement at integer parameters only. The anchor is the variation of the family along its parameter space; its first rung can be tested exactly.

Theorem 4.1 (first-variation anchor). *Let $\Lambda_1^{(n)} = H_n^{(1)} + H_{n+k}^{(1)} + H_{n+l}^{(1)} - 2H_{n-k}^{(1)} - 2H_{n-l}^{(1)} + H_{n+k+l}^{(1)}$ (the n-direction letter form) and $\dot{Q}(n) = \sum_{k,l=0}^{n} T(n,k,l)\, \Lambda_1^{(n)}(n,k,l)$. Let $c_0, \ldots, c_3$ be the certified Brown–Zudilin recurrence coefficients and c_i' their literal polynomial derivatives. Then:*

(1) [VERIFIED: exact $\mathbb{Q}$, $n \le 18$; `eps46_variation.py`]

$$L_{\mathrm{BZ}}(\dot{Q})(n) \;=\; -\sum_{i=0}^{3} c_i'(n)\, Q(n+i). \tag{5}$$

(2) [VERIFIED: exact $\mathbb{Q}$, $n \le 34$] *The k- and l-direction first variations vanish identically: $\sum_{k,l} T \cdot \Lambda_1^{(k)} = \sum_{k,l} T \cdot \Lambda_1^{(l)} = 0$; these are the gauge directions (summation relabelings; the N_1 residue identities).*

(3) [OPEN] *(proof route, identified) The certificate proving $L_{\mathrm{BZ}}(Q) = 0$ is a rational-function identity in n; it therefore holds for the Γ-continued family in continuous n, and* (5) *is its n-derivative, the summation-range variation contributing only at second order because continuation cells vanish doubly. Writing this out against an explicit certificate upgrades (1) to* [PROVED].

Remark 4.2 (the variational tower, with its rung-two obstruction). Iterating, j-th variations should satisfy the j-fold differentiated inhomogeneous tower over the same recurrence, and family-1's Bell weight B_5 is a polynomial in decoupled letter jets, i.e. a $\mathbb{Q}$-combination of mixed fifth partials of the multi-parameter cellular family. Two facts sharpen this proposal, one positive and one an obstruction (`work/VARIATIONAL_TOWER.md`):

(1) Rung one has a *purely discrete* proof route: differentiate the rational certificate identity for $L_{\mathrm{BZ}}(Q) = 0$ and sum over the integer lattice; at first order the continuation cells (double zeros of the cell in the $n-k$, $n-l$ letters) stay dead, so no analytic continuation enters at all.

Moreover Brown–Zudilin's own transfer principle (integrand-level $\mathbb{Q}$-relations descend to the rows separately) licenses the integral-level version of every rung.

(2) [VERIFIED: exact $\mathbb{Q}$, $n \le 15$; `eps47_tower2.py`] The *sum-level* tower fails at rung two in the n-direction: second derivatives revive the continuation cells (a double zero contributes to a second derivative), and the revived layer diverges. Seven of the nine letter directions are safe at all jet orders; the $n-k$, $n-l$ directions are unsafe at order ≥ 2. As tabulated, B_5 requires order-4 jets in the unsafe directions, so the *sum-level* tower cannot reach it; whether the four-dimensional L_4/L_5 null freedom admits a representative with unsafe order ≤ 1 is [OPEN], and the proper home of the higher rungs is in any case the *integral-level* tower $\partial_{v_1 \cdots v_j} I(a)$, where no support issues exist and the transfer principle applies.

If either escape closes the tower, five differentiations plus initial values prove the weight-five attachment $\sum_{k,l} T \cdot B_5 = \frac{33}{4} P_n$ with no certificate search. The compact-versus-Bell comparison (Δ_5) would remain [OPEN]; the sharp low-prime denominator bound would transfer to the B_5 weight by re-running the proved endpoint-residue analysis on it (the programme's "alternate route" completion standard).

5. Applications of the jet calculus

The same machinery, run in reverse (scan $\to$ rational reconstruction $\to$ exact validation), is a discovery instrument for companion formulas. We report four applications; all scans at two unrelated 22-bit primes with identical ranks and verdicts.

5.1. **Rediscovery with controls.** For the Franel family $S = \binom{n}{k}^3$ and the family $\mathbf{D}$: $S = \binom{n}{k}^2\binom{n+k}{n}$, whose companions are known, the pinned linear-atom scan rediscovers reachability and the extracted, rationally reconstructed weights verify exactly over $\mathbb{Q}$ on held-out ranges [VERIFIED: exact $\mathbb{Q}$, $n \le 25$; `eps43.py`]. The discovered weights differ from the published ones by null weights ($\sum_k S \cdot (\text{difference}) = 0$, exact, $n \le 20$) — new equivalent representations. This validates the loop end-to-end on ground truth; a harness defect (an int64 overflow) was caught by the control, illustrating why controls are mandatory.

5.2. **A reachability dichotomy across the fifteen sporadic pairs.** Running the identical protocol on the three sporadic families for which the status paper proves no harmonic-weight fit exists ($\mathbf{B}$, δ, ζ) yields: companion *unreachable* by linear and quadratic curve atoms, at ε^2 and ε^3, in every ζ-graded component [EXCLUDED]. With the two reachable controls this aligns exactly with arithmetic type: reachable = principal character with real $\zeta(2)$-limit; unreachable = non-principal character or no Apéry limit. We state the induced expectation as a conjecture.

Conjecture 5.1. *The ε-jet mechanism on a sporadic cell generates the second solution if and only if the level of the family's modular uniformization is squarefree. (An earlier character-and-limit form of this conjecture was refuted by its discriminating experiment — the Domb family and ε, principal character with real limits and possessing harmonic companions, are jet-unreachable; see the companion paper `papers_out/modular_anchors` and `eps54_domb.py`.)*

5.3. **Closing the unrun fits.** The sporadic status paper listed three full-degree fits as not made (424–1088 columns). All were run (`eps44.py`): families ζ and η over their complete tame alphabets at degree three, and $\mathbf{F}$ at its seven tame arguments, pure and with conductor-matched twisted letters (χ_{-3}, χ_5): *all six inconsistent*, excess 90–173 [EXCLUDED]. Together with the earlier exclusions, the $\sum_k S \cdot \mathfrak{w}$ harmonic ansatz over complete tame alphabets is exhausted for all seven conjectural families. Two diagnostic corrections: ζ *does* possess summand–letter $\mathbb{Q}$-relations at full degree (15-dimensional pure kernel, 36-dimensional twisted), contradicting the earlier "rank = columns" report for ζ; η has rank = columns even at 2530 columns.

5.4. **Annihilation laws: alphabets are quotients.** Extraction of ζ's kernel produces per-cell annihilation identities, e.g.

$$\sum_l S_\zeta(n,k,l)\left(H_k^{(1)} - 2H_l^{(1)} + H_{n-l}^{(1)}\right) \;=\; 0 \qquad \text{for all } 0 \le k \le n \tag{6}$$

and a weight-two companion law [VERIFIED: exact $\mathbb{Q}$, all cells $n \le 25$]; together they generate 12 of the 15 kernel dimensions. Under the inner sum the letters $H_l^{(1)}, H_{n-l}^{(1)}$ are not free — every previous fit for ζ was conducted in a degenerate alphabet. Both laws are single-variable hypergeometric-weighted sums. *Update:* the first law (6) is now [PROVED] for all $0 \le k \le n$, by an order-2 Zeilberger certificate with symbolically verified certificate identities, endpoint vanishing, and unconditional upward induction (`work/z5eps/eps45_zeta_kernel.md`; stated as a Proposition in the companion paper `papers_out/modular_anchors`) — the first proved structural fact about the letters of a conjectural-seven family. The second (weight-two) law remains [VERIFIED: exact $\mathbb{Q}$, $n \le 25$] with a documented obstruction record [OPEN].

5.5. **Modular anchors exactly where jets fail.** The dichotomy of §5.2 predicts that the unreachable families carry structure invisible to binomial letters. A nome computation (recurrence $\to$ MUM Frobenius pair $\to q = t\exp(g/y_0)$; instrument validated on the Apéry $\zeta(3)$ control, which reproduces the classical $\Gamma_0(6)$ eta-quotient parametrization exactly to q^{26}; `eps48_modular_nome.py`) shows that *all three* unreachable families $\mathbf{B}, \delta, \zeta$ possess integer nome expansions — they are modular-parametrized — and identifies ζ's weight-two form exactly:

$$F_\zeta(q) \;=\; \tfrac{1}{8}\left(9E_2(q^9) - E_2(q)\right) \in M_2(\Gamma_0(9)) \qquad \text{[VERIFIED: coefficientwise to } q^{26}\text{]}.$$

$\mathbf{B}$'s weight-one form is identified as well: $F_{\mathbf{B}}(q) = b(-q)$, the Borwein cubic theta $\eta(q)^3/\eta(q^3)$ at $-q$ [VERIFIED: all 27 computed coefficients]. Thus the anchor exists precisely where every binomial-world instrument provably fails, and its letters are q-side objects.

The anchor delivers. Exact boundary bookkeeping shows the second solution of ζ's recurrence satisfies $L(y_B) = t$, and the operator factors through the nome as $L = (P_3(t)/\sigma^3)\cdot F \cdot \theta_q^3 \circ (1/F)$; inverting,

$$B_\zeta(n) \;=\; [t^n]\; F \cdot \theta_q^{-3}\Big(\frac{t\,\sigma^3}{P_3\,F}\Big), \tag{7}$$

an Eichler-type integral in the identified modular data [VERIFIED: coefficientwise exact $\mathbb{Q}$, $n \le 22$, for ζ and for the Apéry $\zeta(3)$ control]. Equation (7) is a companion formula for a family whose complete tame harmonic ansatz is provably empty (§5.3) — the first for any of the conjectural seven, in modular letters. (*Update:* subsequently proved for all n — for all fifteen sporadic families — via the companion inversion and symmetric-square rectification theorems of `papers_out/modular_anchors`; the finite verification stands as an implementation check.) Further, the χ_{-3}-twisted Lucas law for ζ holds with twist exponent $e = 1$ exactly (discriminating primes $5, 11$), consistent with the mod-p shadow of Atkin–Swinnerton-Dyer for the identified $\Gamma_0(9)$ parametrization; the descent problems of the sporadic status paper thus acquire a modular route for ζ. Details and honest limits (nome integrality checked, not proved) in `work/MODULAR_COACTION_PROBE.md`.

6. Limits, falsifiers, and outlook

6.1. **Limits of the negatives.** Curve degrees ≤ 3; direction boxes as stated; $n \le 28$; two 22-bit primes. Deeper curves cannot change Theorem 3.1's top-grade mechanism (Lemma 2.2 is degree-independent), but the finite rank statements are box-dependent as labelled. The sporadic negatives do not exclude constant-term (Gorodetsky-type) representations or rational-function coefficients; those are the recommended next ansätze.

6.2. **Falsifiers.** (i) A pinned curve combination reaching P at any degree would refute the finite part of Theorem 3.1 (the lemmas cannot break). (ii) A weight-three form whose row equals P_n would break the graded-spike input. (iii) First variations failing to anchor in derivative-ladder data would break Remark 4.2's proposal; Theorem 4.1(1) is the surviving first test, and it passed.

6.3. **The mirror side of the $\zeta(5)$ family.** The nome instrument applied to the Brown–Zudilin operator itself (`eps50_z5_nome.py`, `work/Z5_MODULARITY_PROBE.md`) finds: the order-nine operator's indicial structure carries a five-dimensional unipotent block (exponent 0 with multiplicity five — the $(1, 1, 1, 1, 1)$ Calabi–Yau-fourfold-type signature, matching weight five), and the **mirror map is integral with no rescaling**: $t(q) = q - 94q^2 - 591q^3 - 454656q^4 - \cdots$ and $F(q) = 1 + 21q + 1015q^2 + \cdots$ lie in $\mathbb{Z}[[q]]$ [VERIFIED: q^{26}]. (*Correction:* earlier drafts also cited the root-integrality $Q(4t)^{1/2}, Q(8t)^{1/4} \in \mathbb{Z}[[t]]$ as a Dwork-type signal; this is *universal* for integer series — $\sqrt{f(4t)} \in \mathbb{Z}[[t]]$ for any $f \in 1 + t\mathbb{Z}[[t]]$, see `papers_out/half_apery`, Lemma — and is withdrawn as evidence. Whether the block is operator-level Sym^4 remains [OPEN]; no global order-5 right factor of t-degree ≤ 20 exists.) The integral mirror map, and the integral Yukawa-type datum computed subsequently, are the genuine arithmetic signatures: mirror-type signatures consonant with the measured Frobenius spike $\mathrm{diag}(1, p^3, p^5)$, one floor above the elliptic-modular sporadics; see the companion paper `papers_out/modular_anchors` (conjectural tier) for the precisely scoped statements. We record the induced expectation as the *mirror form of the bridge*: in the block's q-coordinate the companion rows should be coefficient sequences of q-side Eichler-type objects (as (7) realises for the sporadic ζ), and (2) the t-shadow of a q-side identity; together with Theorem 3.1 this says the weight-five-seeing instrument should be built from the block's own coordinates rather than binomial letters.

6.4. **Outlook.** The natural home of the tower is the moduli-space family: parameter derivatives are the Gauss–Manin connection, the recurrence is the contiguity direction, and (5) is their compatibility. The jet-coaction dictionary (jets as de Rham shadows of the motivic coaction, with the sporadic dichotomy as Galois equivariance) is conjectural but has paid rent: it retrodicts the one-step (unipotent) shape of (5) — the inhomogeneity couples to Q alone — and it makes the falsifiable prediction that second variations anchor in $\mathrm{span}\{Q, \dot{Q}\}$ only, which the tower's second rung will test. The immediate mathematical targets are: a parametric certificate for the cellular family; the second rung of the tower; and WZ certificates for (6). On the first target, Brown's recent Mellin-transform framework [5] provides, in its appendix for the $\zeta(2)/\mathcal{M}_{0,5}$ case, exactly the required object in a toy: explicit contiguity matrices $M_i(s)$, *rational in continuous parameters*, transporting an anchored period vector across the parameter lattice; differentiating the rational transport chain anchors parameter-derivatives in closed form. Constructing the analogous matrices for the $\zeta(5)/\mathcal{M}_{0,8}$ family is the identified one-time cost of the tower programme (see `work/BROWN26_MINING.md`).

7. Provenance and data

All scripts in `work/z5eps/` of the programme repository, with logs: `eps40_gate.py` (realizability obstruction); `eps41.py`, `eps42.py` (curve scans, Theorem 3.1 finite part); `eps43.py` (+ `eps43_*.pkl`: controls and dichotomy); `eps44.py` (the unrun fits, §5.3); `eps46_variation.py` (Theorem 4.1); memos `work/CURVE_BLINDNESS.md`, `work/EPS_DISCOVERY.md`, `work/sharp12_lowprimes/BRIDGE_C`
All exact arithmetic is `Fraction`-based; every modular verdict was recomputed at a second, unrelated prime with identical rank and verdict.

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